Airbnb Price Prediction Report — New York City

Generated on 2026-08-25 by the automated pipeline (`make all`). Source: InsideAirbnb.

Executive summary

This report analyzes the **New York City** Airbnb market using 21,328 cleaned listings and 26 listing features. A machine-learning model was trained to predict nightly prices and then evaluated on a **holdout set** — listings it had never seen before.

The best model was **XGBoost**. On the holdout set it achieved:

- **Average error (MAE): $116** — a typical prediction is off by about this much.
- **RMSE: $572** — like MAE, but weighs large errors more heavily.
- **MAPE: 48%** — the average error as a share of the actual price.
- **R^2: 0.92** — the share of price variation the model explains (0–1).
- **44%** of predictions were within +/-25% of the true price.

How to read this report

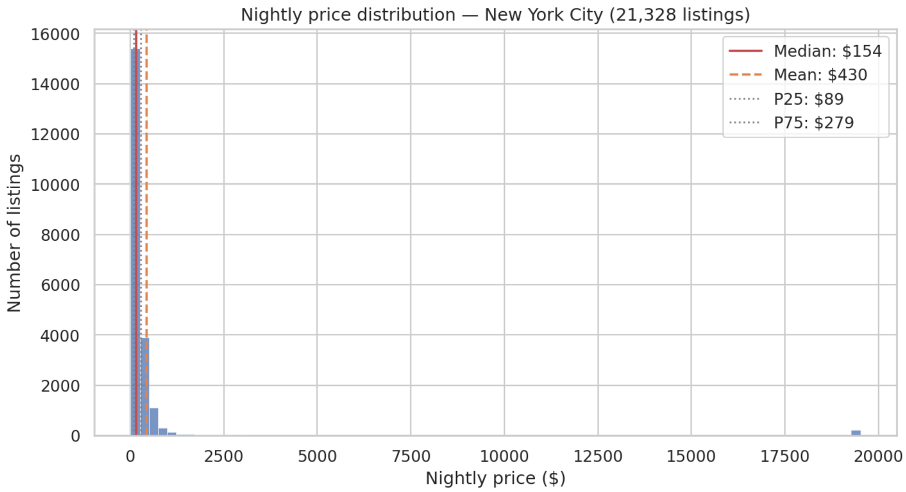
- **MAE (mean absolute error)** — the average difference between predicted and actual prices, in dollars. Lower is better; it's the easiest number to grasp ("off by about $X on average").
- **RMSE (root mean squared error)** — similar to MAE but penalizes large mistakes more. If RMSE is much higher than MAE, a few big errors are inflating it.
- **MAPE (mean absolute percentage error)** — the same idea as MAE, but expressed as a percentage of the actual price, so it can be compared across cities.
- **R^2** — how much of the variation in prices the model explains (0 = no better than guessing the average; 1 = perfect).
- **Holdout set** — the 10% of listings the model never saw while training. All accuracy numbers come from this set, so they are an honest estimate of real-world performance.
- **Top features** — the listing attributes the model leans on most when setting a price.

Data

- Source: InsideAirbnb (latest available snapshot)
- Listings after cleaning: **21,328** — median nightly price **$154**, mean **$430**
- Features used by the model: **26**
- Train / validation / holdout split: **14,929 / 3,839 / 2,560** listings (70/15/10, random seed 42)

Market overview

Price distribution



- Median nightly price: **$154**; the middle 50% of listings fall between **$89** and **$279**. - The mean (**$430**) sits well above the median — a minority of expensive listings pull the average up, so the **median is the better guide** to a typical price.

Price by room type

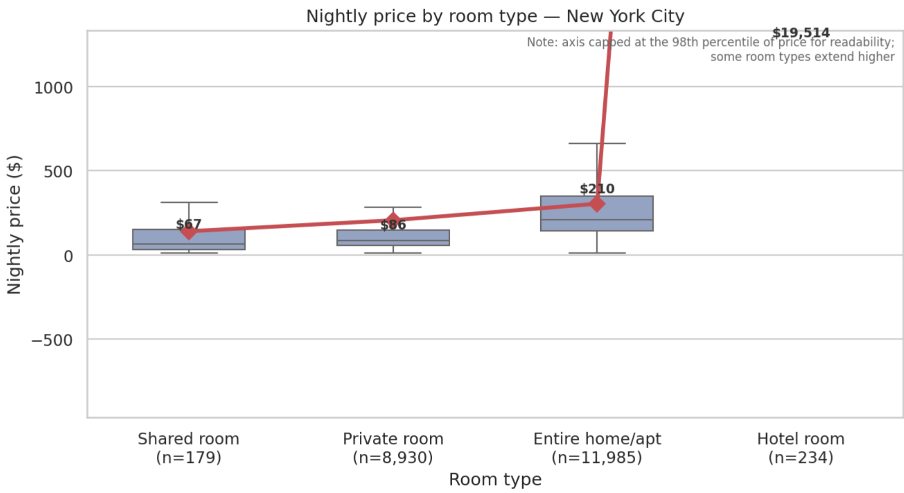


Figure 1: price_by_room_type

Room type	Listings	Median price	Mean price
Hotel room	234	$19514	$15668
Entire home/apt	11,985	$210	$304
Private room	8,930	$86	$206
Shared room	179	$67	$141

Price by number of bedrooms

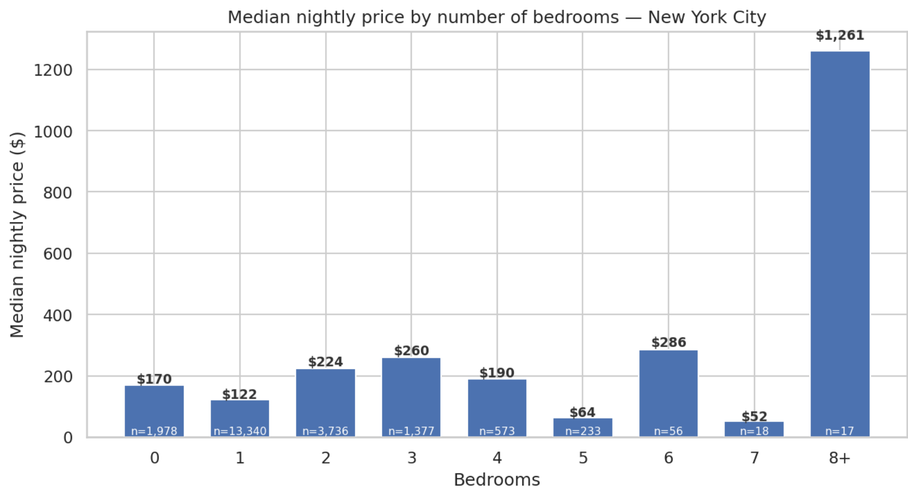


Figure 2: price_by_bedrooms

Feature correlations

Model accuracy on the holdout set

Metrics for the chosen model (**XGBoost**) on 2,560 unseen listings:

Metric	Value	What it means
MAE	$116	Average prediction is off by this much
RMSE	$572	Errors, with big misses weighted more
MAPE	48%	Average error as a % of the true price
R^2	0.92	Share of price variation explained

Accuracy in practice: 20% of predictions were within +/-10% of the true price, 44% within +/-25%, and 70% within +/-50%.

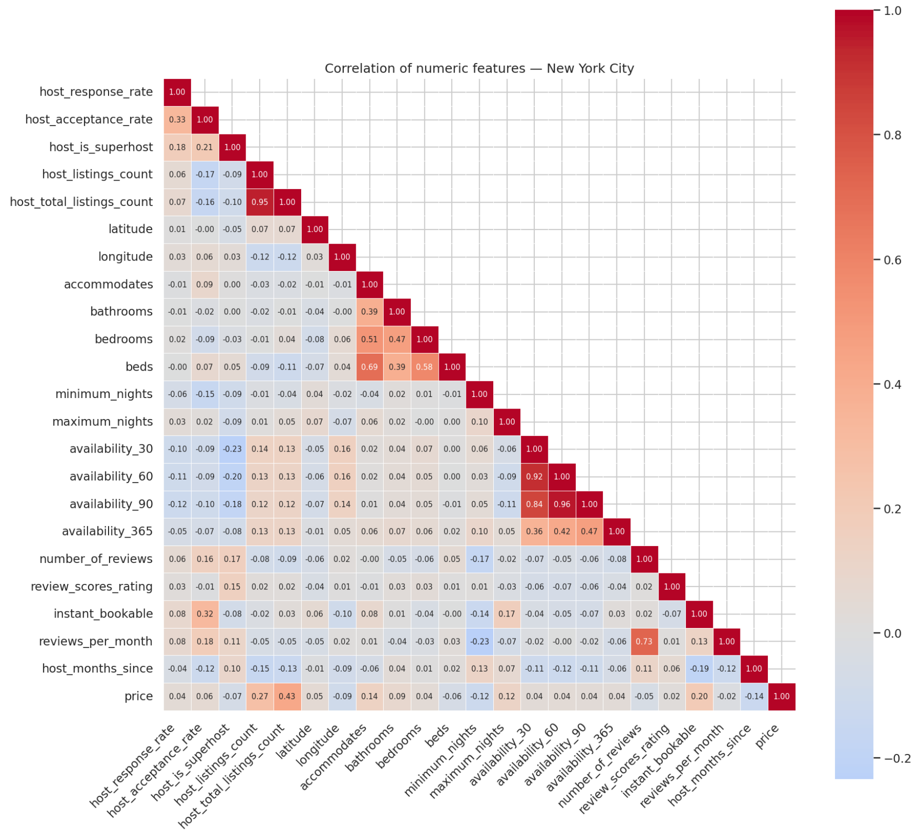
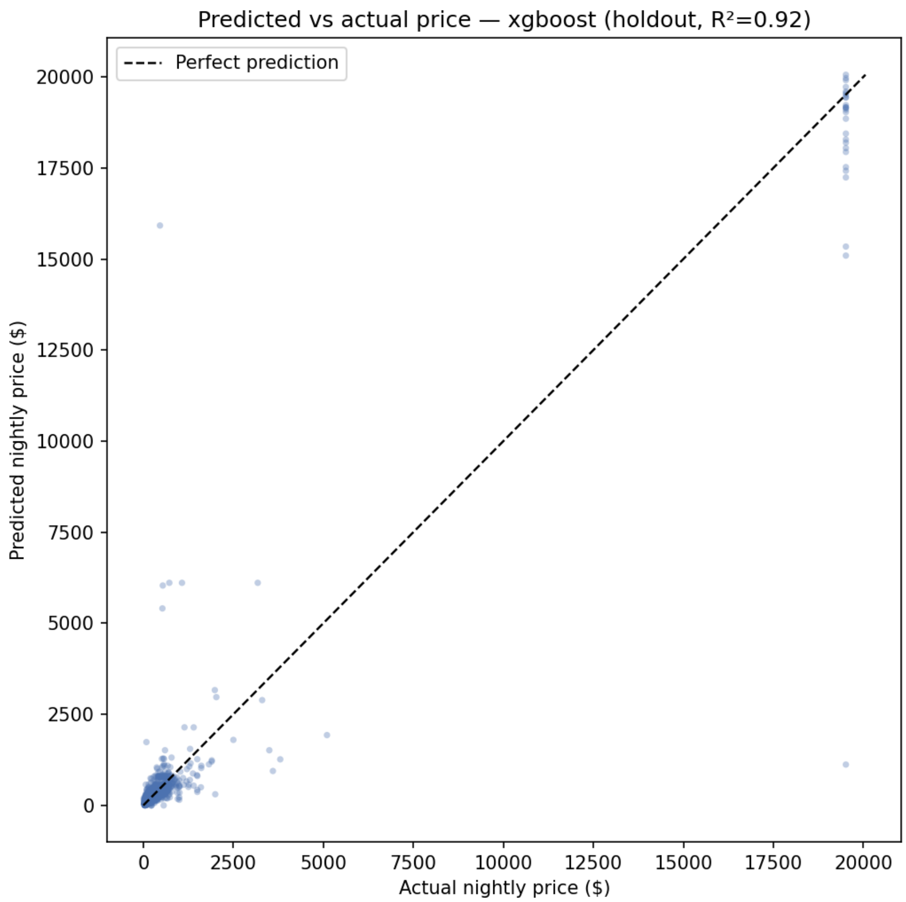
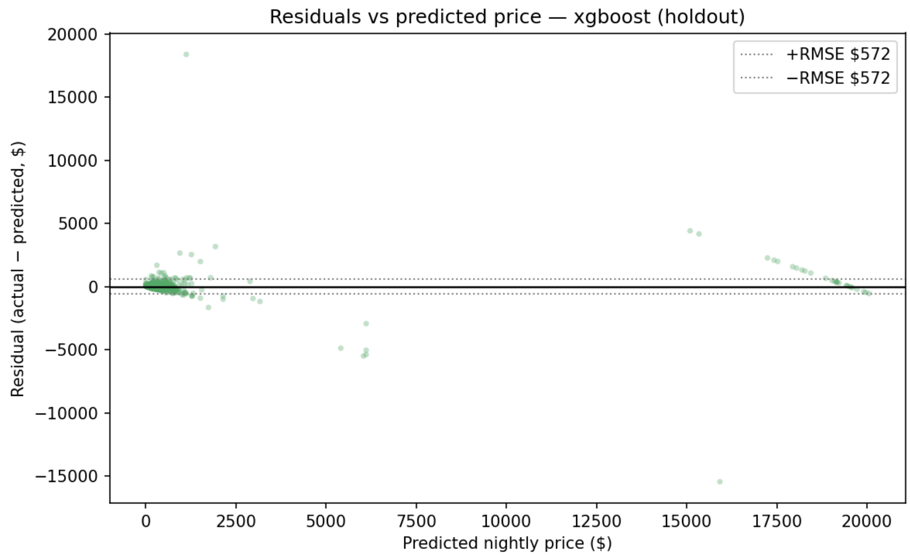


Figure 3: corr__heatmap



Points close to the dashed line are accurate predictions; points above it mean the model under-predicted.



Residual = actual - predicted. A cloud centered on the zero line means errors are balanced rather than systematically too high or too low.

Models & results

Model	Validation RMSE
Random Forest	$713
XGBoost	$667
TabPFN (capped sample, not directly comparable)	$956

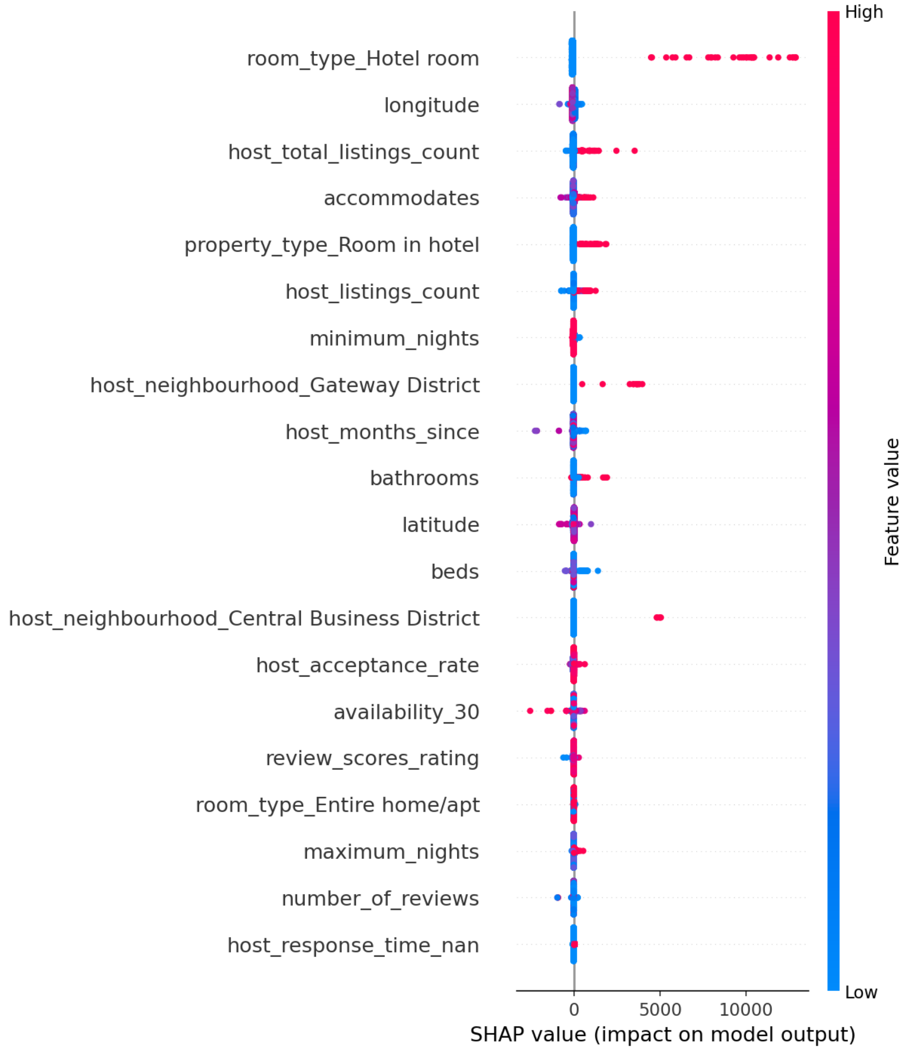
The model with the lowest validation RMSE was then scored once on the holdout set — see the accuracy section above for its final numbers.

Top features

The listing attributes that influence price the most (importance = how much the model relies on them):

Feature	Importance	Meaning
room_type_Hotel room	0.400	the room type (entire home vs private/shared room) (specifically Hotel room)

Feature	Importance	Meaning
host_response_time_within an hour	0.134	how quickly the host responds (specifically within an hour)
host_neighbourhood_Gateway District	0.084	the host's neighbourhood (specifically Gateway District)
host_neighbourhood_Central Business District	0.049	the host's neighbourhood (specifically Central Business District)
host_months_since	0.027	how long the host has been hosting (months)
host_neighbourhood_Airport North	0.025	the host's neighbourhood (specifically Airport North)
property_type_Room in hotel	0.022	the property type (apartment, house, etc.) (specifically Room in hotel)
beds	0.021	the number of beds
host_neighbourhood_Clearwater Beach	0.019	the host's neighbourhood (specifically Clearwater Beach)
minimum_nights	0.018	the minimum nights required



SHAP summary: each dot is one listing; colour shows the feature's value (red = high, blue = low) and position shows whether it pushed the predicted price up (right) or down (left).

Prediction examples

A few real holdout listings with the model's prediction:

Listing	Actual	Predicted	Error	Error %
accommodates=1, bedrooms=1, room_type=Private room	$77	$167	$90	116.9%
accommodates=1, bedrooms=1, room_type=Private room	$117	$118	$1	0.9%
accommodates=8, bedrooms=4, room_type=Entire home/apt	$347	$402	$55	15.8%
accommodates=1, bedrooms=0, room_type=Entire home/apt	$241	$199	$42	17.4%
accommodates=1, bedrooms=1, room_type=Private room	$44	$6	$38	86.3%
accommodates=2, bedrooms=1, room_type=Entire home/apt	$212	$235	$23	10.7%
accommodates=2, bedrooms=1, room_type=Private room	$116	$120	$4	3.3%
accommodates=2, bedrooms=1, room_type=Private room	$85	$80	$5	6.4%

Recommendations

- The best-performing model was **XGBoost** (holdout RMSE $572, MAE $116, R^2 0.92).
- On validation, **XGBoost** beat Random Forest by 6% on RMSE ($667 vs $713). A simple ensemble or blending the two may squeeze out further gains.
- Pricing is most strongly driven by **room_type_Hotel room, host_response_time_within an hour, host_neighbourhood_Gateway**

District — i.e. room_type_Hotel room (the room type (entire home vs private/shared room) (specifically Hotel room)); host_response_time_within an hour (how quickly the host responds (specifically within an hour)); host_neighbourhood_Gateway District (the host's neighbourhood (specifically Gateway District)). Hosts can use these levers when setting nightly rates.

- On average, predictions are off by about **48%** of the actual price (median error 30%), and **44%** of predictions landed within +/-25% of the true price.
- Accuracy caveat: the holdout RMSE of $572 is **371% of the median listing price ($154)**, so predictions for typical listings are expected to be off by roughly +/-$572.
- Next steps: add richer features (neighbourhood_cleansed, amenities, review scores breakdowns), train on a log-transformed target to tame high-end outliers, run a deeper hyperparameter search, and extend the pipeline to more cities.

Limitations

- The model only "sees" the features listed in the config — it cannot capture neighbourhood charm, amenities, photos, or seasonality.
- Extreme high-end prices are clipped at the 99th percentile during cleaning, so predictions at the very top of the market are less meaningful.
- The accuracy numbers are averages: individual predictions can be much better or much worse.
- This report is generated automatically from the pipeline artifacts and reflects the data as of 2026-08-25.

Automated pipeline: fetch → preprocess → EDA → train → evaluate → report. Code in `src/`, config in `configs/base.yaml`.